In [1]:
```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
```

In [2]:
```
# Load Dataset
df = pd.read_csv("ign_bracket_competition (1).csv")
```

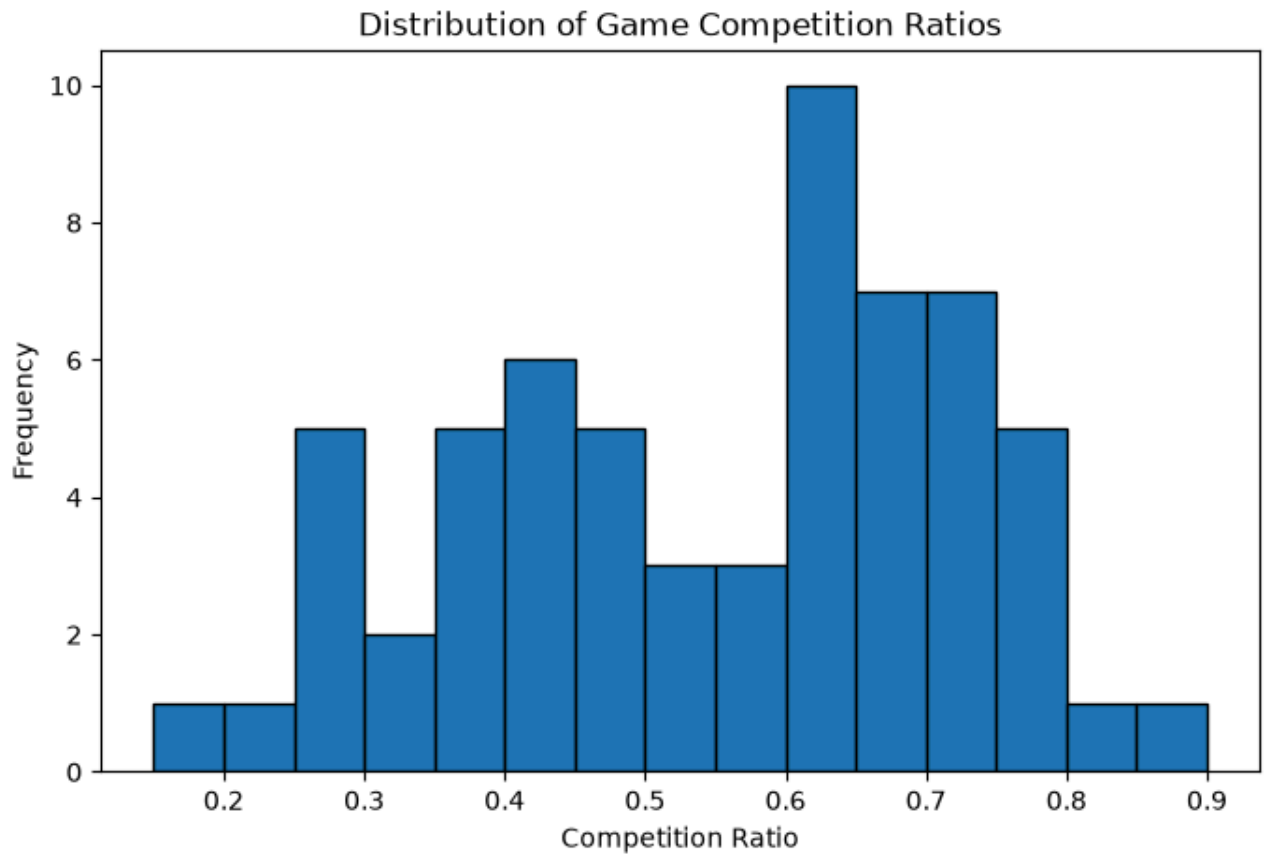
In [3]:
```
# Display Dataset Information
print("Dataset Shape:", df.shape)
display(df.head())
print(df.info())
```

```
Dataset Shape: (62, 4)
```

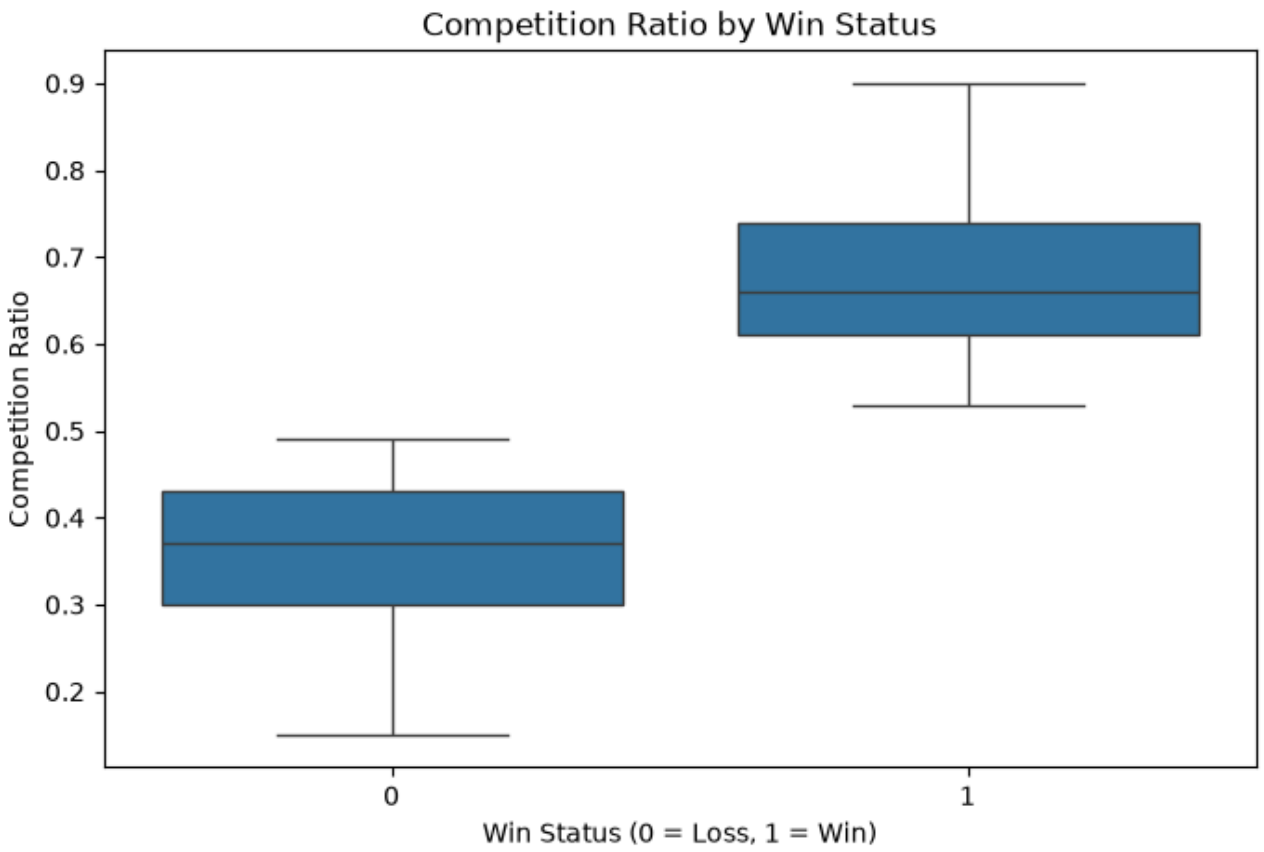
	Game	Opponent	Ratio	Win
0	Red Dead Redemption	Borderlands 2	0.79	1
1	Grand Theft Auto: San Andreas	Mario Kart 8 Deluxe	0.73	1
2	Star Wars: Knights of the Old Republic	Tony Hawk Pro Skater 2	0.55	1
3	Fallout: New Vegas	League of Legends	0.74	1
4	World of Warcraft	Fortnite	0.62	1

```
<class 'pandas.DataFrame'>
RangeIndex: 62 entries, 0 to 61
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Game      62 non-null     str
 1   Opponent  62 non-null     str
 2   Ratio     62 non-null     float64
 3   Win       62 non-null     int64
dtypes: float64(1), int64(1), str(2)
memory usage: 2.1 KB
None
```

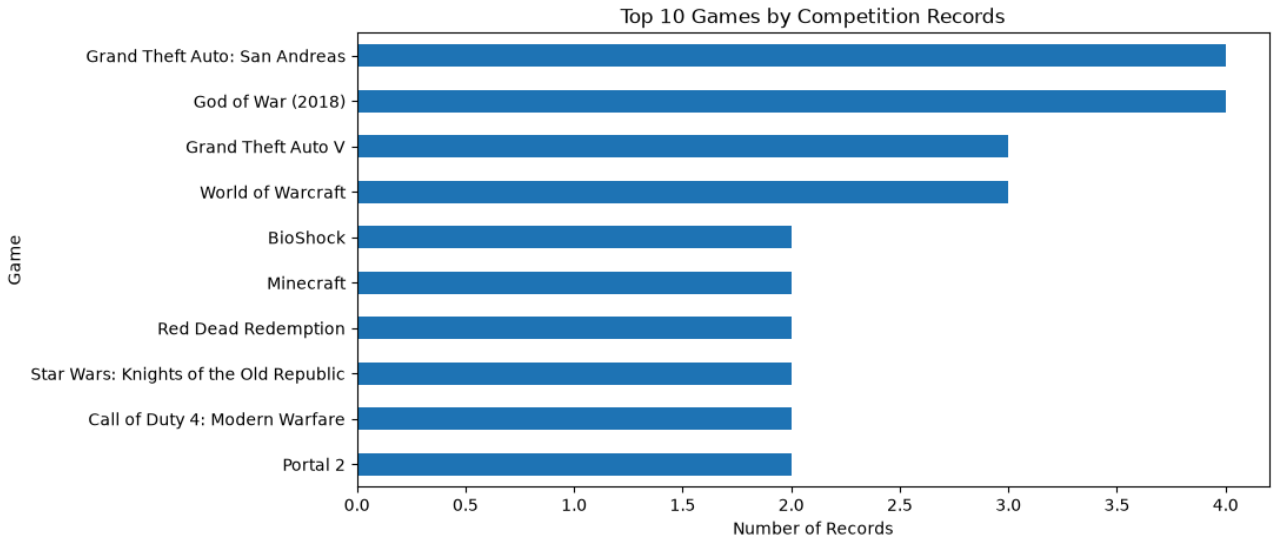
In [4]:
```
# 1. Distribution of Competition Ratio
plt.figure(figsize=(8, 5))
plt.hist(df['Ratio'], bins=15, edgecolor='black')
plt.title('Distribution of Game Competition Ratios')
plt.xlabel('Competition Ratio')
plt.ylabel('Frequency')
plt.show()
```



In [5]:
```
# 2. Competition Ratio by Win Status
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='Win', y='Ratio')
plt.title('Competition Ratio by Win Status')
plt.xlabel('Win Status (0 = Loss, 1 = Win)')
plt.ylabel('Competition Ratio')
plt.show()
```

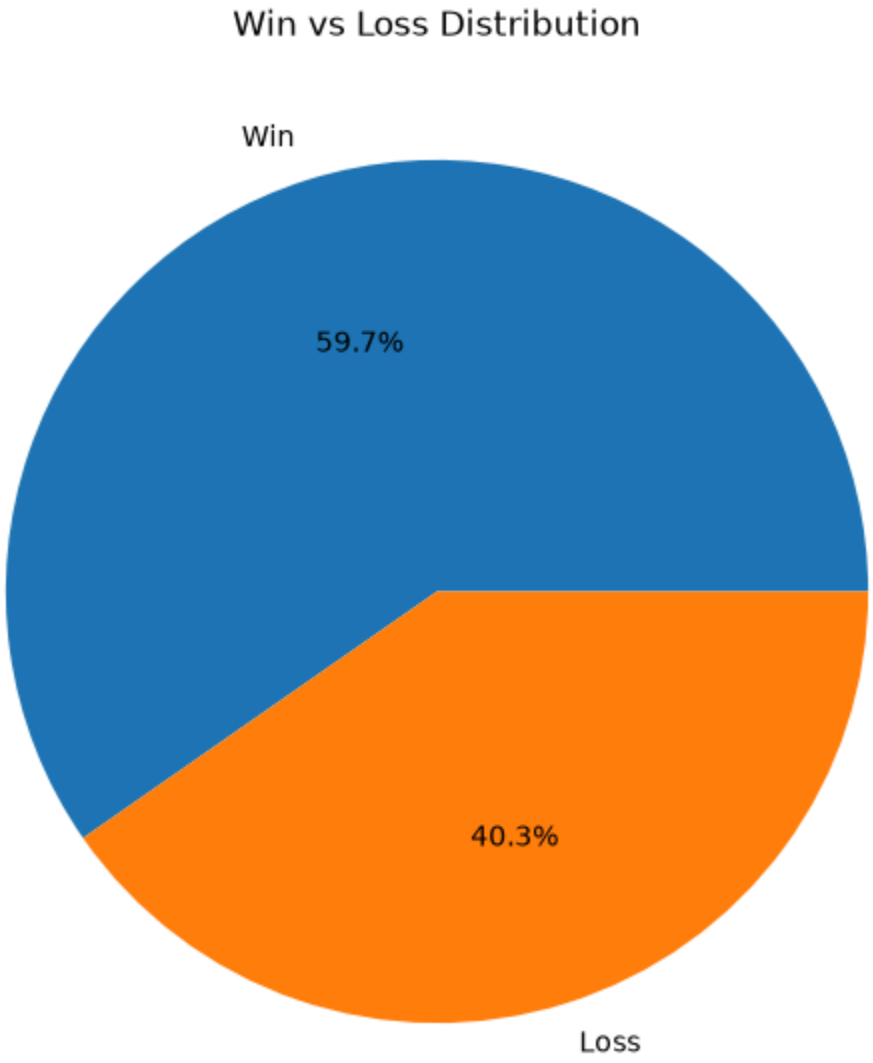


In [6]:
```
# 3. Top 10 Games by Number of Appearances
plt.figure(figsize=(10, 5))
df['Game'].value_counts().head(10).sort_values().plot(
    kind='barh'
)
plt.title('Top 10 Games by Competition Records')
plt.xlabel('Number of Records')
plt.ylabel('Game')
plt.show()
```



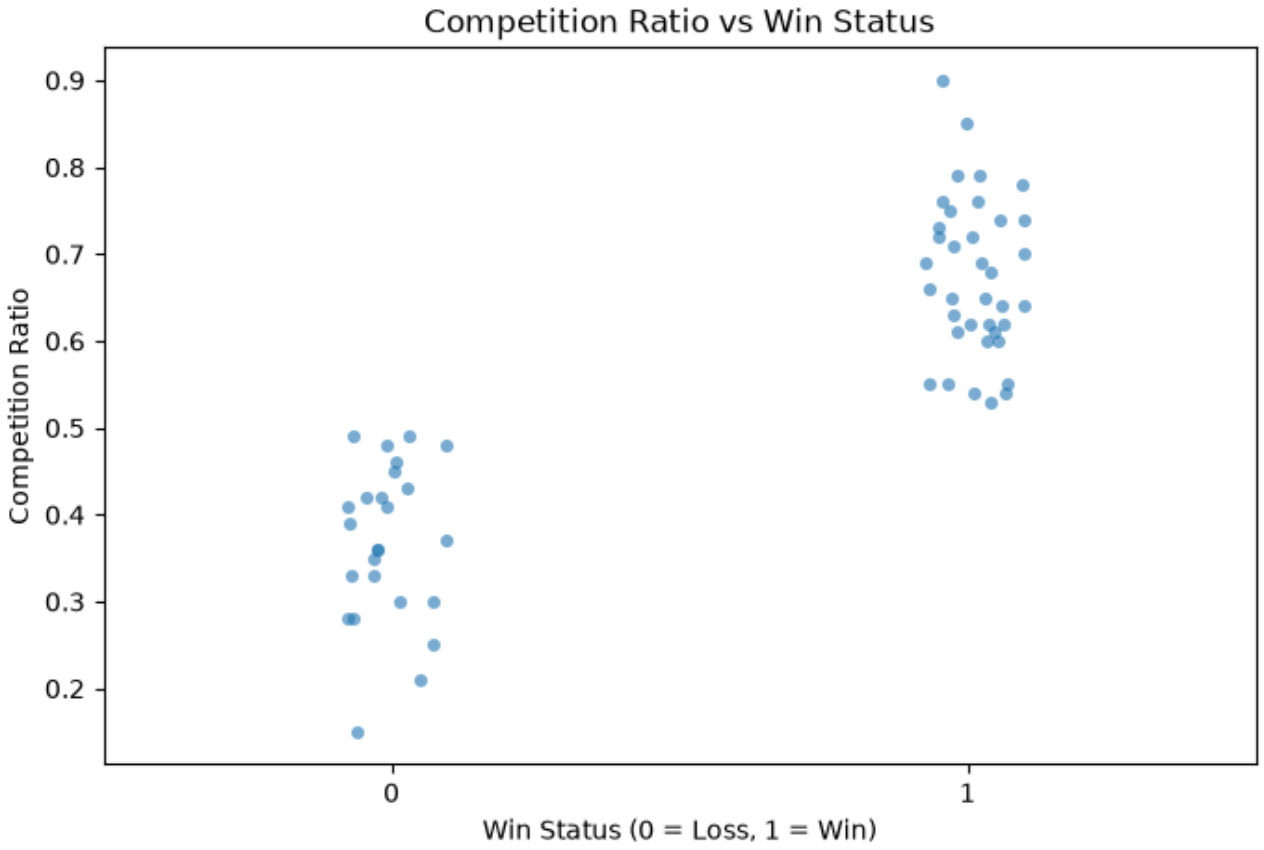
In [7]:
```
# 4. Win vs Loss Distribution
df['Win'].value_counts().rename(
    index={0: 'Loss', 1: 'Win'}
).plot(kind='pie', autopct='%1.1f%%', figsize=(7, 7))

plt.title('Win vs Loss Distribution')
plt.ylabel('')
plt.show()
```

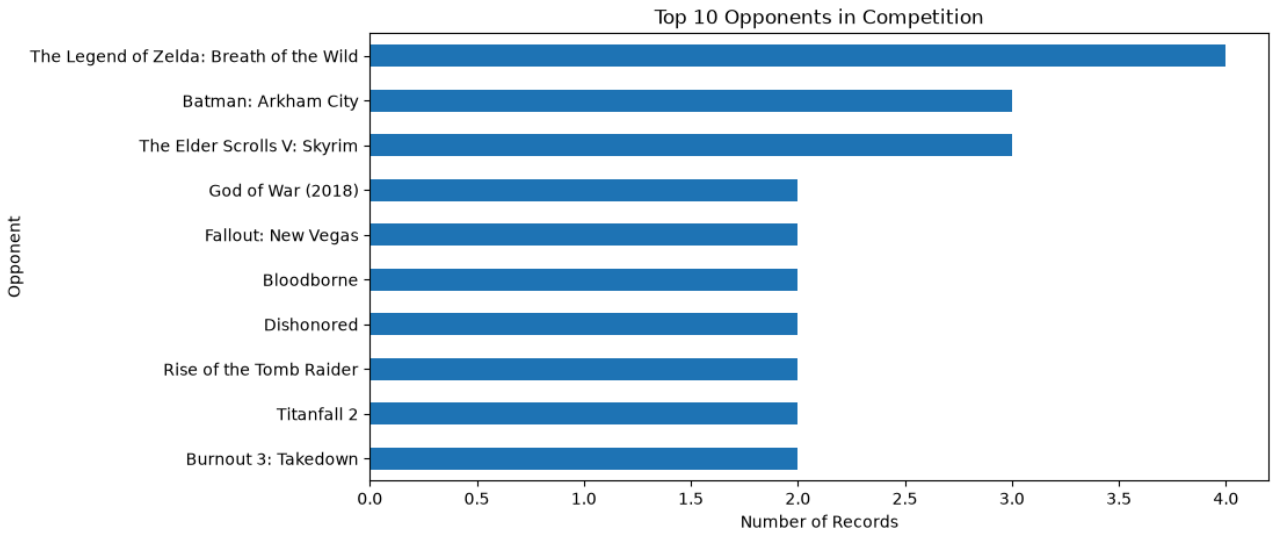


In [8]:
```
# 5. Competition Ratio vs Win Status
plt.figure(figsize=(8, 5))
sns.stripplot(
    data=df,
    x='Win',
    y='Ratio',
    alpha=0.6,
    jitter=True
```

```
)
plt.title('Competition Ratio vs Win Status')
plt.xlabel('Win Status (0 = Loss, 1 = Win)')
plt.ylabel('Competition Ratio')
plt.show()
```



In [9]:
```
# 6. Top 10 Opponents by Number of Appearances
plt.figure(figsize=(10, 5))
df['Opponent'].value_counts().head(10).sort_values().plot(
    kind='barh'
)
plt.title('Top 10 Opponents in Competition')
plt.xlabel('Number of Records')
plt.ylabel('Opponent')
plt.show()
```


Top 10 Opponents in Competition
The Legend of Zelda: Breath of the Wild
Batman: Arkham City
The Elder Scrolls V: Skyrim
God of War (2018)
Fallout: New Vegas
Bloodborne
Dishonored
Rise of the Tomb Raider
Titanfall 2
Burnout 3: Takedown
Opponent
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Number of Records

In [ ]: